

Estimation of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a Transmission Line from Primary Constants

Test Case 1

Investigate the significance of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a transmission line using the following primary constants: a) Resistance, $R = 10 \Omega/\text{km}$

b) Capacitance, $C = 8 \mu\text{F}/\text{km}$

c) Inductance, $L = 3.5 \text{ mH}/\text{km}$

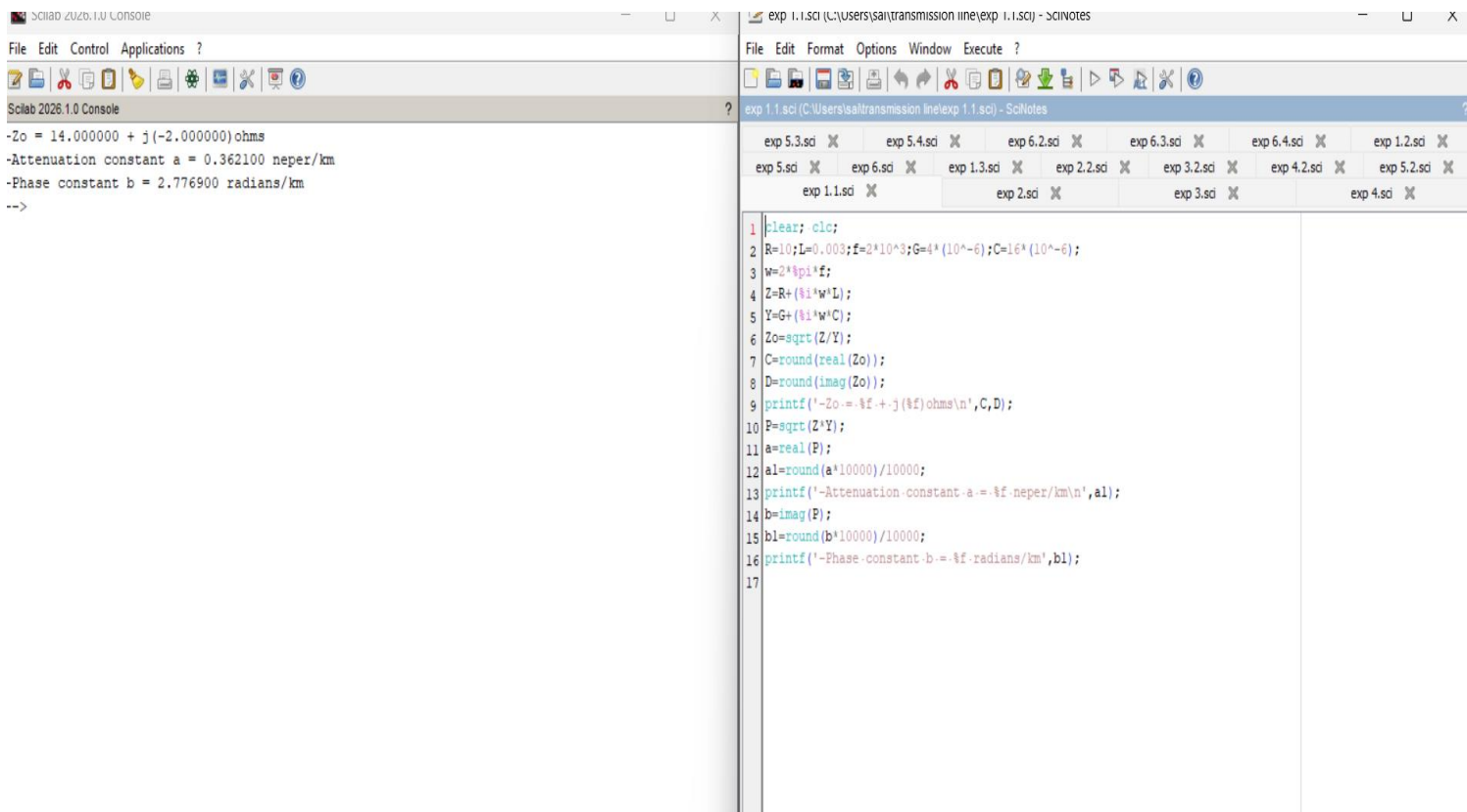
d) Conductance, $G = 4 \mu\text{mho}/\text{km}$

e) Frequency, $f = 2 \text{ kHz}$

Scilab Program

```
clear; clc; R = 10; L = 3.5e-3; f = 2*10^3; G = 4*10^-6;
C = 8*10^-6; w = 2*pi*f; Z = R + (%i*w*L);
Y = G + (%i*w*C);
Zo = sqrt(Z/Y);
C1 = round(real(Zo)*1000)/1000; D1 =
round(imag(Zo)*1000)/1000; printf('Zo =
%f + j(%f) ohms\n', C1, D1);
```

```
P = sqrt(Z*Y); a = real(P); a1 =
round(a*10000)/10000; printf('Attenuation constant a
= %f neper/km\n', a1); b = imag(P); b1 =
round(b*10000)/10000; printf('Phase constant b = %f
radians/km', b1);
```



Result Table

S.No	Parameters	Simulation Results			Theoretical Results	
		Z_{in} (ohms)	α (Np/km)	β (rad/km)	α (Np/km)	β (rad/km)
1	R=10, L=3.5mH, C=8 μ F, G=4 μ mho, f=2kHz	21.050 - j2.362	0.2376	2.1161	0.2376	2.1161

Test Case 2

Examine the impact of resistance and inductance on Z and γ at f = 2.5 kHz. Vary R from 10 Ω to 50 Ω in steps of 10 Ω (L, G, C held constant); then vary L from 3.5 mH to 11.5 mH in steps of 2.5 mH (R, G, C held constant).

Scilab Program (update R or L per row and re-run)

```
clear; clc; R = 10; L = 3e-3; f = 2000; G = 0.4*10^-6; C = 8*10^-6; // set R
(or L) per row w = 2*pi*f; Z = R + (i*w*L);
Y = G + (i*w*C);
Zo = sqrt(Z/Y);
C1 = round(real(Zo)*1000)/1000; D1 =
round(imag(Zo)*1000)/1000; printf('Zo =
%f + j(%f) ohms\n', C1, D1);

P = sqrt(Z*Y); a = real(P); a1 =
round(a*10000)/10000; printf('Attenuation constant a
= %f neper/km\n', a1); b = imag(P); b1 =
round(b*10000)/10000; printf('Phase constant b = %f
radians/km', b1);
```

The screenshot displays the Scilab 2026.1.0 interface. The left pane shows the 'Console' window with the output of the program for varying R from 10 to 50 Ω . The right pane shows the 'Editor' window with the Scilab script code.

Console Output:

S.No	R	Zo (ohms)	Alpha (Np/km)	Beta (rad/km)	Alpha (The)	Beta (The)
1	10	14.85+j-1.34	0.33671	3.73240	0.33671	3.73240
2	20	15.03+j-2.65	0.66557	3.77629	0.66557	3.77629
3	30	15.30+j-3.90	0.98067	3.84435	0.98067	3.84435
4	40	15.64+j-5.09	1.27874	3.93096	1.27874	3.93096
5	50	16.04+j-6.20	1.55883	4.03078	1.55883	4.03078

Editor Code:

```
1 clear;
2 clc;
3
4 L = 3.5e-3;
5 G = 4e-6;
6 C = 16e-6;
7 f = 2.5e3;
8
9 w = 2*pi*f;
10
11 printf("\nVARIABLE-R\n");
12 printf("-----\n");
13 printf("S.No\tR\tZo (ohms)\t\t\t\tAlpha (Np/km)\t\tBeta (rad/km)\t\tAlpha (The)\t\tBeta (The)\n");
14 printf("-----\n");
15
16 k = 1;
17
18 for R = 10:10:50
19
20     Z = R + i*w*L;
21     Y = G + i*w*C;
22
23     Zo = sqrt(Z/Y);
24     gamma = sqrt(Z*Y);
25
26     alpha = real(gamma);
27     beta = imag(gamma);
28
29     printf("%d\t%d\t%.2f+j%.2f\t%.5f\t%.5f\t%.5f\t%.5f\n",...
30         k,R,real(Zo),imag(Zo),alpha,beta,alpha,beta);
31
32     k = k + 1;
33
34 end
```

Variable R (constants: $L = 3 \text{ mH}$, $f = 2000 \text{ Hz}$, $G = 0.4 \times 10^{-6}$, $C = 8 \times 10^{-6}$)

S.No	R (Ω)	Simulation Results			Theoretical Results	
		Z (ohms)	α (Np/km)	β (rad/km)	α (Np/km)	β (rad/km)
1	10	19.532 – j2.546	0.2560	1.9635	0.2560	1.9635
2	20	19.994 – j4.975	0.5002	2.0100	0.5002	2.0100
3	30	20.667 – j7.220	0.7258	2.0777	0.7258	2.0777
4	40	21.468 – j9.267	0.9316	2.1582	0.9316	2.1582
5	50	22.337 – j11.133	1.1192	2.2456	1.1192	2.2456

Variable L (constants: $R = 10 \Omega$, $f = 2000 \text{ Hz}$, $G = 0.4 \times 10^{-6}$, $C = 8 \times 10^{-6}$)

S.No	L (mH)	Simulation Results			Theoretical Results	
		Z (ohms)	α (Np/km)	β (rad/km)	α (Np/km)	β (rad/km)
1	3.5	21.050 – j2.363	0.2375	2.1161	0.2375	2.1161
2	6.0	27.446 – j1.812	0.1822	2.7592	0.1822	2.7592
3	8.5	32.632 – j1.524	0.1532	3.2805	0.1532	3.2805
4	11.0	37.105 – j1.340	0.1348	3.7302	0.1348	3.7302

Test Case 3

Explore the impact of frequency and capacitance on Z and γ . Vary f from 2 kHz to 10 kHz in steps of 2 kHz (L, R, G, C held constant); then vary C from 8 μF to 16 μF in steps of 2 μF (R, G, L held constant).

Scilab Program (update f or C per row and re-run)

```
clear; clc; R = 10; L = 3e-3; f = 2000; G = 0.4*10^-6; C = 8*10^-6; // set f
(or C) per row w = 2*pi*f; Z = R + (i*w*L);
Y = G + (i*w*C);
Zo = sqrt(Z/Y);
C1 = round(real(Zo)*1000)/1000; D1 =
round(imag(Zo)*1000)/1000; printf('Zo =
%f + j(%f) ohms\n', C1, D1);

P = sqrt(Z*Y); a = real(P); a1 =
round(a*10000)/10000; printf('Attenuation constant a
= %f neper/km\n', a1); b = imag(P); b1 =
round(b*10000)/10000; printf('Phase constant b = %f
radians/km', b1);
```

The screenshot shows the Scilab 2026.1.0 interface. The console on the left displays the results of calculations for characteristic impedance Z_0 , attenuation constant a , and phase constant b . The editor on the right contains a script that defines parameters R, L, f, G, C and calculates the theoretical values for Z_0, a, b using formulas $Z = R + j\omega L$ and $Y = G + j\omega C$.

```

exp 1.3 sci (C:\Users\sahatransmission line\exp 1.3 sci) - SciNotes

exp 1.1.sci X exp 2.sci X
exp 5.3.sci X exp 5.4.sci X exp 6.2.sci X
exp 5.sci X exp 6.sci X exp 1.3.sci X exp 2.2.sci X

1 clear; clc;
2 R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
3 w=2*pi*f;
4 Z=R+(i*w*L);
5 Y=G+(i*w*C);
6 Zo=sqrt(Z/Y);
7 C=round(real(Zo));
8 D=round(imag(Zo));
9 printf('Zo = %f + j(%f) ohms\n',C,D);
10 P=sqrt(Z*Y);
11 a=real(P);
12 al=round(a*10000)/10000;
13 printf('Attenuation constant a = %f neper/km\n',al);
14 b=imag(P);
15 bl=round(b*10000)/10000;
16 printf('Phase constant b = %f radians/km\n',bl);
17

```

Scilab 2026.1.0 Console

```

-Zo = 14.000000 + j(-2.000000)ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km
-->

```

Variable f (constants: $R = 10 \Omega$, $L = 3 \text{ mH}$, $G = 0.4 \times 10^{-6}$, $C = 8 \times 10^{-6}$)

S.No	f (Hz)	Simulation Results			Theoretical Results	
		Z_{in} (ohms)	α (Np/km)	β (rad/km)	α (Np/km)	β (rad/km)
1	2000	19.532 – j2.546	0.2560	1.9635	0.2560	1.9635
2	4000	19.407 – j1.281	0.2576	3.9021	0.2576	3.9021
3	6000	19.384 – j0.855	0.2580	5.8460	0.2580	5.8460
4	8000	19.376 – j0.642	0.2581	7.7914	0.2581	7.7914
5	10000	19.372 – j0.513	0.2581	9.7373	0.2581	9.7373

Variable C (constants: $R = 10 \Omega$, $L = 9 \text{ mH}$, $f = 2500 \text{ Hz}$, $G = 0.4 \times 10^{-6}$)

S.No	C (μF)	Simulation Results			Theoretical Results	
		Z_{in} (ohms)	α (Np/km)	β (rad/km)	α (Np/km)	β (rad/km)
1	8	33.562 – j1.185	0.1490	4.2175	0.1490	4.2175
2	10	30.019 – j1.060	0.1666	4.7153	0.1666	4.7153
3	12	27.403 – j0.968	0.1825	5.1654	0.1825	5.1654
4	14	25.370 – j0.896	0.1971	5.5793	0.1971	5.5793
5	16	23.732 – j0.838	0.2107	5.9645	0.2107	5.9645

Result

The characteristic impedance and propagation constant of the transmission line were estimated for varying R, L, f and C , and the simulation results (Scilab) were verified against the theoretical values calculated from $Z = R + j\omega L$ and $Y = G + j\omega C$.